Antioxidant Activity of Aqueous Extract of Pericarp of *Luffa aegyptiaca* (Sponge Gourd)

Curcurbitaceae Family

A project report submitted to the Department of Pharmacy, Khwaja Yunus Ali University in partial fulfillment of the requirements for the degree of Bachelor of Pharmacy

Submitted By:

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Semester: Summer 2024

Department of Pharmacy

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Submission Date: 17th April, 2025

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Dedicated With Deepest Gratitude To

My Beloved Parents

and

My Esteemed Supervisor,

Md. Shihab Uddin Sohag

Declaration

I, Fahmida Anam Mithila, hereby declare that the dissertation entitled **"Antioxidant Activity of Aqueous Extract of Pericarp of *Luffa aegyptiaca*(Sponge Gourd)"**submitted by myself to the Department of Pharmacy, Khwaja Yunus Ali University, in the partial fulfillment of the requirement for the award of the degree Bachelor of Pharmacy is a complete record of original research work carried out by me during 2024, under the supervision and guidance of **Md. Shihab Uddin Sohag**, Lecturer, Department of Pharmacy, Khwaja Yunus Ali University and the dissertation have not formed the basis for awarding any other degree/diploma/fellowship or other similar titles to any candidate of any university.

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Department of Pharmacy

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This is to certify that the project titled **"Antioxidant Activity of Aqueous Extract of Pericarp of *Luffa aegyptiaca*(Sponge Gourd)"** was submitted to the Department of Pharmacy, Khwaja Yunus Ali University, in the partial fulfillment of the requirement for the degree of Bachelor of pharmacy was carried out by **Fahmida Anam Mithila (**ID: 0622030211091001) in 2025, under my supervision and guidance. The project has not formed the basis for awarding any other degree/ diploma/fellowship or any other similar title to any university candidate.

Md. Shihab Uddin Sohag

Lecturer

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I am grateful to Almighty Creator for giving me the patience and ability to finish this project. I wish to express my gratitude to my family for their unwavering inspiration, financial assistance, and moral support.

An individual never completes a project. It is more than just a compilation of people's thoughts, recommendations, evaluations, contributions, and labor.Without guidelines, it cannot be finished.

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Abstract

Background: Known by many names as Egyptian cucumber or sponge gourd, *Luffa aegyptiaca* is used extensively in traditional medicine around the world, but especially in tropical and subtropical countries. Its fruit, peel, and seeds have long been used by indigenous people to cure a variety of illnesses, including fever, gastrointestinal issues (including enteritis), inflammatory diseases, and localized edema. In addition to its medicinal uses, the plant is praised for its abundant nutritional profile, which includes vitamins, minerals, and bioactive substances including phenolic acids and flavonoids. In this study we aim to evaluate antioxidant activity of aqueous extract of pericarp of *Luffa aegyptiaca.*

Methods: The DPPH (2,2-diphenyl-1-picrylhydrazyl) radical scavenging experiment & ABTS (2,2′-azinobis-3-ethylbenzthiazolin-6-sulfonic acid) was used to assess the plant's antioxidant capability.

Results & Discussion: The extract from the pericarp assay of the *Luffa aegyptiaca* showed good antioxidant activity when compared to ascorbic acid in the DPPH free radical scavenging assay; the extract's IC_{50} value was determined to be 485.843±88.902 µg/ml, whereas the standard solution's(Ascorbic Acid) IC_{50} value was 9.287±2.495µg/ml. The ABTS assay extract's IC_{50} value was 200.964±45.202 µg/mL, compared to the standard (ascorbic acid) IC_{50} value of 10.139±3.195 µg/mL.

Conclusion: Findings indicate that *Luffa aegyptiaca* pericarp extract possesses promising antioxidant properties under the tested conditions. Further exploration is warranted to uncover additional bioactive potentials and validate its applicability in pharmaceutical or nutraceutical contexts.

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List of Abbreviations

- µg: Microgram Per Mili-Liter
- HPTLC: High-Performance Thin-Layer Chromatography
- DPPH: 1, 1- Diphenyl-2-Picrylhydrazyl.
- IC50: Half-Maximal Inhibitory Concentration
- WHO: World Health Organization
- LC-MS: Liquid Chromatography-Mass Spectrometry
- NCEs : New Chemical Entities
- LLE: Liquid-Liquid Extraction
- SPE: Solid-Phase Extraction

Chapter One:
Introduction

1. Introduction

Plants are fundamental to life on Earth, serving as sources of food, medicine, oxygen, shelter, and ecological stability. They play a vital role in human nutrition by providing essential nutrients such as carbohydrates, proteins, fats, vitamins, and minerals. Additionally, plants are indispensable in medicine, offering therapeutic compounds like alkaloids, flavonoids, and terpenoids that are used to treat conditions ranging from cancer to diabetes (Albahri et al., 2023). The integration of plants into human environments—through green spaces or indoor gardening—has been shown to enhance air quality, reduce stress, and improve overall well-being (Stepansky et al., 2023).

Medicinal plants have been used for millennia in traditional healing systems such as Ayurveda and Traditional Chinese Medicine. These systems emphasize the holistic use of plant-derived remedies for physical and mental health. Modern science has increasingly recognized the therapeutic potential of these plants, leading to their incorporation into contemporary healthcare practices (Schulz et al., 2001; Morgan et al., 2002). For example, bioactive compounds from plants like Taxus brevifola (paclitaxel) and Catharanthus roseus (vinblastine and vincristine) have been developed into life-saving drugs for cancer treatment (Richard et al., 2022).

Among medicinal plants, *Luffa aegyptiaca* stands out due to its diverse applications. Traditionally consumed as a vegetable, its mature fruit is used as a bath sponge while its leaves are utilized in treating conjunctivitis. The plant contains bioactive compounds such as saponins, flavonoids, tannins, and cardiac glycosides that exhibit antimicrobial, anti-inflammatory, and antioxidant properties (Du et al., 2006; Panicker et al., 2020). Its seeds are rich in oil (44.8%) composed primarily of polyunsaturated fatty acids and proteins (39%), making them valuable for livestock feed and human nutrition (Prakas et al., 2013).

Pharmacological studies have demonstrated the potential of *Luffa aegyptiaca* extracts in treating respiratory diseases like asthma and bronchitis. The plant's antifungal activity against pathogens such as Fusarium oxysporum further underscores its medicinal significance (Muthumani et al., 2012). Its ability to act as a purgative and diuretic has also been documented in traditional medicine systems across Asia and Africa (Panicker et al., 2020).

The global surge in demand for herbal medicines has driven research into the safety, efficacy, and bioactivity of plant-derived compounds. Reverse pharmacology—where traditional knowledge guides laboratory validation—has emerged as a cost-effective approach for drug discovery. This method has facilitated the development of standardized herbal formulations with reduced toxicity (Wang et al., 2023; Du et al., 2006). Collaborative efforts among taxonomists, pharmacologists, and biochemists continue to explore new applications for plants like *Luffa aegyptiaca*, aiming to address unmet medical needs through innovative therapies (Rahul et al., 2024).

1.1 Medicinal Plant

The World Health Organization (WHO) defines medicinal plants as those containing substances in one or more of their organs that can be used for therapeutic purposes or serve as precursors for drug synthesis. Medicinal plants have been utilized for centuries in traditional medicine, and their significance persists in modern healthcare due to their bioactive compounds, such as alkaloids, terpenoids, phenolics, and glycosides, which exhibit diverse pharmacological properties (Petrovska, 2012; Mamadalieva et al., 2008). These compounds are the basis for many pharmaceutical drugs, including aspirin, quinine, and paclitaxel (Dar et al., 2023).

Globally, approximately 80% of the population relies on medicinal plants for primary healthcare needs. In Bangladesh, where access to modern healthcare is limited to only 40% of the population, traditional plant-based remedies play a critical role in addressing health challenges. Bangladesh's rich biodiversity includes thousands of medicinal plant species used in Unani and Ayurvedic systems of medicine (Saha, 2023). With over 50,000 medicinal plants documented worldwide, these plants represent a significant resource for drug discovery and development (Frontiers Plant Science, 2023).

1.2 Classification of medicinal plant

Medicinal plants, with their diverse therapeutic applications, are classified based on various criteria such as the part used, habit, habitat, and therapeutic value. Approximately 80,000 higher plant species are known to possess medicinal properties, and around 5,000 of these species have specific therapeutic uses (Joy et al., 1998). This classification system helps in understanding their medicinal potential and organizing them for research and application.

Based on Part Used

Medicinal plants are categorized based on the specific parts utilized for therapeutic purposes:

Whole Plant: Examples include *Boerhaavia diffusa* and *Phyllanthus niruri*, which are used for various ailments like liver disorders and kidney stones.

Root: Dasamula is traditionally used in Ayurvedic medicine for respiratory disorders and inflammation.

Stem: Plants like *Tinospora cordifolia* and *Acorus calamus* are known for their immunomodulatory and digestive health benefits.

Bark: *Saraca asoca* is widely used in gynecological treatments.

Leaf: *Lawsonia inermis* (henna) and Aloe vera have applications in wound healing and skin care.

Flower: *Biophytum sensitivum* is used for its antioxidant properties, while *Mimusops elenji* is applied in traditional medicine for respiratory issues.

Fruit: Solanum species are used for their anti-inflammatory effects.

Seed: Datura stramonium is utilized for its antispasmodic properties.

Based on Habit

Habit refers to the physical growth form of plants:

Grasses: *Cynodon dactylon* is used for its diuretic properties.

Sedges: *Cyperus rotundus* has applications in digestive health.

Herbs: *Vernonia cinerea* is known for its anti-inflammatory effects.

Shrubs: Solanum species are widely studied for antimicrobial activities.

Climbers: *Asparagus racemosus* is valued for its adaptogenic properties.

Trees: *Azadirachta indica* (neem) has broad-spectrum antimicrobial applications.

Based on Habitat
Medicinal plants thrive in distinct ecological environments:

Tropical: *Andrographis paniculata*, commonly known as "king of bitters," is used in liver protection and immune enhancement therapies.

Sub-tropical: Mentha arvensis is employed in digestive health treatments.

Temperate: *Atropa belladonna*, a source of atropine, has applications in ophthalmology and pain management.

Based on Therapeutic Value
Medicinal plants are grouped according to their pharmacological effects:

Antimalarial: *Cinchona officinalis*, a source of quinine, and *Artemisia annua*, known for artemisinin production, are widely used against malaria.

Anticancer: Alkaloids from *Catharanthus roseus* and paclitaxel from *Taxus baccata* are key components in chemotherapy drugs.

Antidiabetic: Plants like *Momordica charantia* help regulate blood glucose levels.

Antiviral: Extracts from *Acacia catechu* exhibit antiviral activity against pathogens like herpes simplex virus.

Antibacterial: *Plumbago indica* contains compounds effective against bacterial infections.

Hepatoprotective: Silymarin from *Silybum marianum* protects the liver from toxins and oxidative stress.

1.3 Use of Plants as a Therapeutic Tool
Plants have been a cornerstone of medicine, providing therapeutic compounds from various parts such as bark, roots, leaves, and fruits. These bioactive compounds are utilized in anti-inflammatory, analgesic, CNS stimulant and depressant, cardiovascular, antidiabetic, and anticancer medications. However, not all active constituents are beneficial; some may exhibit toxicity, requiring exclusion from therapeutic use. The complexity of plants, which contain multiple phytoconstituents like alkaloids, flavonoids, and terpenoids, poses challenges in isolating pure compounds for medicinal purposes (Mukherjee et al., 2007).

Traditionally, single active ingredients are extracted, refined, and standardized for drug development. Advanced techniques such as DNA fingerprinting, high-performance thin-layer chromatography (HPTLC), and liquid chromatography-mass spectrometry (LC-MS) are now employed to separate and identify multiple active constituents in herbal formulations (Pandey et al., 2017). These methods ensure the safety and efficacy of plant-based medicines while addressing the complexity of their chemical composition.

1.4 The Value of Certain Plants as Medicine

Medicinal plants are highly valued for their therapeutic properties due to the presence of bioactive compounds that exert significant biological effects. These compounds make plants useful as stimulants, poisons, hallucinogens, or medicines. The key bioactive chemicals contributing to the medicinal importance of plants include: i. Alkaloids: These nitrogen-containing compounds exhibit pain-relieving, addictive, or toxic effects and are essential in treating various ailments. For instance, morphine from the opium poppy (*Papaver somniferum*) is widely used as an analgesic (Dar et al., 2023; Niazi & Monib, 2024). ii. Glycosides: Known for their role in heart stimulation, purgative effects, and sexual health enhancement, glycosides like digoxin from Digitalis species are critical in managing cardiovascular diseases (Dar et al., 2023). iii. Tannins: These polyphenolic compounds are effective against gastrointestinal disorders such as diarrhea and ulcers and also aid in wound healing and skin disease treatment due to their astringent properties (Niazi & Monib, 2024). iv. Volatile/Essential Oils: Found in aromatic plants, these oils enhance digestion, improve appetite, and exhibit antiseptic and insecticidal properties. For example, thyme essential oil is known for its antibacterial and antioxidant effects (Dar et al., 2023; Sohani, 2021). v. Vitamins and Minerals: Fruits and vegetables are rich sources of essential vitamins (e.g., A, C) and minerals (e.g., potassium, magnesium), which contribute to overall health and are widely used in herbal remedies (Sohani, 2021).

1.5 The indirect role of medicinal herbs in contemporary synthetic medicine

Medicinal plants have significantly influenced modern synthetic drug development in the following ways:

a. Novel Chemical Structures

Bioactive compounds from plants often inspire chemists to design semi-synthetic derivatives. For example, paclitaxel from *Taxus brevifolia* led to the creation of synthetic analogs like docetaxel with enhanced anticancer activity (Newman & Cragg, 2020).

b. Structural Modifications

Plant-derived molecules are chemically modified to improve therapeutic effects. Aspirin, for instance, was synthesized by modifying salicylic acid from willow bark (Salix alba), enhancing its anti-inflammatory properties (Atanasov et al., 2021).

c. Development of Analogues

Synthetic analogues of plant compounds are engineered for superior efficacy. Artemisinin from Artemisia annua inspired derivatives like artesunate, which are more effective against malaria (Patra et al., 2021).

1.6 The Value of Medicinal Plants in the Development of New Drugs

Medicinal plants play a crucial role in drug discovery, offering a rich source of bioactive compounds that serve as templates for developing new drugs. Current research integrates botanical, phytochemical, biological, and molecular techniques to explore their potential against diseases such as cancer, HIV/AIDS, Alzheimer's, and malaria. Notable plant-derived drugs like arteether, galantamine, and paclitaxel have been introduced into modern medicine or are undergoing clinical trials (Balunas & Kinghorn, 2005). Data shows that natural products or their derivatives account for approximately 52% of all New Chemical Entities (NCEs) developed in the last two decades, underscoring their significance in pharmaceutical innovation (WHO, 2025). Furthermore, with 80% of the global population relying on traditional medicine for primary healthcare, medicinal plants remain indispensable in both developed and developing countries

(Niazi & Monib, 2024). The pharmaceutical industry benefits from single-entity plant-derived drugs like atropine, digoxin, and vincristine, which treat severe illnesses. Although some older compounds have declined in use due to better alternatives, advancements in drug development are projected to increase the market share of plant-based single-entity drugs (Balunas & Kinghorn, 2005).With rising healthcare costs and growing interest in natural remedies, medicinal plants continue to drive innovation in modern drug discovery while addressing global health challenges.

1.7 The Future of Medicinal Plants

The future of medicinal plants is promising, with over half a million plant species worldwide, many of which remain unexplored for their medicinal properties. These plants hold immense potential for addressing current and future therapeutic needs. Medicinal plants have been integral to human culture, influencing traditions, ceremonies, and healthcare systems (Hosseinzadeh et al., 2015). Modern drug discovery continues to benefit from medicinal plants. Aspirin, derived from willow bark, and other plant-based drugs like artemisinin and paclitaxel exemplify their significance. Additionally, food crops such as garlic exhibit medicinal effects, highlighting the overlap between nutrition and therapy (Choudhary et al., 2015). The therapeutic effects of medicinal plants are attributed to their secondary metabolites, which exhibit diverse chemical structures and biological activities. Advances in natural product chemistry, including improved isolation and characterization techniques, have expanded the scope of bioactive compound discovery. Genomics and proteomics are also aiding in identifying active compounds and biosynthetic pathways in plants (Clark, 1996; González-Burgos & Gómez-Serranillos, 2025).The World Health Organization (WHO) recognizes the importance of traditional medicine and has developed guidelines for the cultivation, processing, and quality control of medicinal plants. With rising healthcare costs, many governments are promoting the use of traditional medicines as affordable alternatives to expensive pharmaceuticals (WHO, 2025).

Chapter Two:
Plant Preview

2.1 Plant Description

The sponge gourd (*Luffa aegyptiaca*), also known as Egyptian cucumber or Vietnamese luffa, is an annual climbing vine from the Cucurbitaceae family. It is native to South and Southeast Asia and has been cultivated for centuries for its versatile uses. The plant features five-angled stems with tendrils that aid in climbing, and its leaves are broad, ovate, and lobed. The fruits, which can grow up to 61 cm in length, are initially green and smooth but become fibrous as they mature, forming a dense network of cellulose fibers surrounded by a hard shell (Purseglove, 1976; Shah et al., 1980). These fibers are commonly used as natural sponges after processing (CABI Compendium, 2019).

Luffa gourds thrive in well-drained soil with full sun exposure and good air circulation. They are typically grown on sturdy trellis systems to prevent fruit rot and deformities caused by contact with the ground. In commercial cultivation, raised beds with black polyethylene mulch and drip irrigation are recommended to optimize growth in cooler climates. Regular pruning and pollination management, such as placing beehives nearby during flowering, are crucial for high yields (Davis & De Courley, 1994).

The plant has significant medicinal applications. Its fruits are used in traditional Ayurvedic practices to treat conditions such as skin disorders, intestinal worms, and inflammatory diseases. For instance, the fresh juice of the fruit is used to induce purgation in cases of ascites and constipation, while the paste of its leaves is applied to reduce swelling (Easy Ayurveda, 2016). Additionally, phytochemical studies reveal that *Luffa aegyptiaca* contains bioactive compounds such as saponins, flavonoids, oleanolic acid, and lucyosides A-H. These compounds exhibit antimicrobial, antioxidant, anti-inflammatory, and hypolipidemic properties (Agubosi et al., 2022; Rahul et al., 2024).

The nutritional value of *Luffa aegyptiaca* further enhances its appeal. The plant is rich in dietary fiber, vitamins (including vitamin C), minerals like potassium and magnesium, and bioflavonoids that contribute to its health benefits. Studies have also highlighted its potential use in industrial applications such as biofilters and sound-absorbing materials due to the unique properties of its fibers (Bal et al., 2004; Iqbal & Edyvean, 2004).



Figure 1: *Luffa aegyptiaca*

2.2 Botanical Description

2.2.1 Taxonomy

Scientific name/ Botanical name: *Luffa aegyptiaca*
Family: Curcurbitaceae
Kingdom: Plantae
Order: Cucurbitales
Genus: Luffa
Species: aegyptiaca

2.2.2 Geographical Distribution

Luffa cylindrica, commonly referred to as sponge gourd, is a subtropical plant that thrives in warm summer temperatures and requires a long frost-free growing season, particularly when cultivated in temperate regions. It is an annual climbing vine that produces cylindrical fruits with a fibrous vascular system, making it an ideal summer crop (Heiser & Schilling, 1988).

The exact native range of *Luffa cylindrica* is difficult to pinpoint due to its extensive history of cultivation. However, Indo-Burma is widely regarded as the center of diversity for this species. The plant has been cultivated for centuries across tropical regions of Asia and Africa and has naturalized in various parts of the world, including East Africa and Pacific islands (Heiser & Schilling, 1988).

Today, the primary commercial production regions include China, Korea, India, Japan, and Central America. These countries have optimized cultivation techniques to meet the growing demand for luffa sponges in both domestic and industrial markets. Additionally, Brazil has emerged as a significant producer of luffa products in recent years (CABI Compendium, 2019; Mazali & Alves, 2005).

Naturalized populations of *Luffa cylindrica* are found in disturbed areas such as forest edges and agricultural lands. In East Africa, for example, the plant has become naturalized in Kenya and Tanzania and is considered invasive in parts of Uganda. Its adaptability to various climates and soil types has allowed it to thrive in environments ranging from sea level to elevations of up to 1,800 meters.

2.2.3 Synonnym

Luffa aegyptica (lapsus)

Momordica luffa L.

Luffa cylindricaM.Roem.

Luffa pentandraRoxb.

Momordicacylindrica L.

2.2.4 Geographical Distribution

Luffa aegyptiaca, commonly known as sponge gourd or smooth luffa, is an annual subtropical vine cultivated for its versatile fruit. The plant is widely believed to have originated in tropical Asia, with Indo-Burma recognized as the center of diversity. Historical accounts suggest its domestication began in Asia and spread to China around 600 AD and Egypt during the Middle Ages (Purseglove, 1976; Mansfeld, 2001). Today, it is cultivated globally across tropical and subtropical regions, including major commercial production areas such as China, India, Korea, Japan, Central America, and Brazil (Oboh & Aluyor, 2009; CABI Compendium, 2019).

The young fruit of *Luffa aegyptiaca* is consumed as a vegetable in tropical Asia, while the mature fibrous fruit is utilized for industrial purposes such as scrubbing bath sponges and filters. The plant's adaptability has allowed it to naturalize in diverse environments ranging from Africa to Southeast Asia and South America. It thrives in warm climates with long frost-free growing seasons and performs best in fertile soils with adequate sunlight (Seshadri & More, 2009; Davis & DeCourley, 1994).

In addition to its economic significance, *Luffa aegyptiaca* plays a role in traditional medicine and sustainable agriculture. Its seeds can be used to extract edible oil, while the residual meal serves as animal feed or fertilizer. The plant's widespread cultivation highlights its importance as both a food crop and an industrial resource (Joshi et al., 2004; Chabert-Llompart, 2020).

2.2.5 Medicinal Uses

Extraction and Toxicity Studies

Methanolic extraction of *Luffa aegyptiaca* fruits was performed using dried, pulverized material in a Soxhlet apparatus, yielding a 15.5% extract (w/w) stored at 2–8°C for subsequent studies. Acute oral toxicity tests in rats (100–2000 mg/kg) revealed no mortality or adverse effects over 72 hours, indicating a high safety profile for short-term use (Abdel-Salam et al., 2018; Alagbe, 2019).

Hypolipidemic and Hepatoprotective Effects

In hypercholesterolemic rabbit models, methanolic fruit extracts (300 mg/kg/day) significantly reduced serum total cholesterol (29%), triglycerides (52%), and LDL-C (22%), while increasing HDL-C (38%). Similar lipid-lowering effects were observed in diabetic rats treated with *L. aegyptiaca* decoctions, which also normalized LDL-C/HDL-C and TC/HDL-C ratios, critical markers for cardiovascular risk. Liver homogenates from treated animals showed reduced lipid accumulation, suggesting enhanced hepatic lipid metabolism (Thayyil et al., 2011; Nnam & Okorie, 2024).

Mechanisms and Phytochemical Basis

The hypolipidemic activity of *Luffa aegyptiaca* is attributed to bioactive compounds such as saponins, flavonoids, and oleanolic acid derivatives identified in phytochemical analyses. These compounds inhibit cholesterol synthesis and enhance bile acid excretion. Additionally, L. aegyptiaca extracts demonstrated hepatoprotective effects by lowering serum AST and ALT levels in toxin-induced liver injury models, preserving hepatic structural integrity (Giwa et al., 2025; Liu et al., 2019).

Antidiabetic Potential

Methanolic extracts exhibited dose-dependent hypoglycemic effects in alloxan-induced diabetic rats, linked to β-cell regeneration and improved insulin sensitivity. Fermented Luffa formulations further reduced insulin resistance and inflammation in high-fat diet models (Shikano et al., 2019; Zhang et al., 2019).

Chapter Three:
Literature Review

3. Phytochemical Investigation

The phytochemical analysis involved standardized tests to identify key bioactive compounds. For alkaloid detection, the extract was treated with 1% HCl, boiled, and filtered, followed by reactions with Dragendorff's reagent (orange-red precipitate), Mayer's reagent (creamy-white precipitate), and Wagner's reagent (reddish-brown precipitate) (Sahira Banu & Cathrine, 2015; Biology Discussion, 2025). Steroids were identified using the Libermann-Burchard test (blue-green coloration with acetic anhydride and sulfuric acid) and the Salkowski test (reddish-brown interface with chloroform and sulfuric acid) (Orient Journal of Chemistry, 2022; Sahira Banu & Cathrine, 2015). Tannins were confirmed via the ferric chloride test (green-black precipitate) and the lead acetate test (white gelatinous precipitate) (NepJOL, 2025; Sahira Banu & Cathrine, 2015). Proteins were detected using Millon's test (brick-red precipitate with mercuric nitrate) and verified by the biuret test (violet coloration with NaOH and copper sulfate) (NepJOL, 2025; Sahira Banu & Cathrine, 2015).

3.1 Cytotoxic Activity Test

The cytotoxic potential of the extract was evaluated using the brine shrimp lethality bioassay (*Artemia salina*). Brine shrimp eggs were hatched in artificial seawater (38 g/L salinity) at 25°C under constant aeration for 48 hours to obtain mature nauplii (Rahman et al., 2024). Stock solutions of the extract were prepared in dimethyl sulfoxide (DMSO) and serially diluted in seawater to achieve final concentrations of 1, 5, 10, 20, 50, 100, 200, and 500 µg/mL. Control vials contained equivalent DMSO volumes to rule out solvent toxicity (Solis et al., 1993). Ten nauplii were transferred to each vial using a Pasteur pipette, and mortality was recorded after 24 hours.

The lethality percentage was calculated as:
Lethality (%) = (Number of dead nauplii / Total nauplii) × 100

The LC_{50} (median lethal concentration) was determined via linear regression analysis of dose-response curves (Meyer et al., 1982).

3.2 Anthelmintic Assay

The anthelmintic activity of the extract was evaluated using adult earthworms (*Pheretima posthuma*), selected due to their anatomical and physiological resemblance to human intestinal roundworm parasites (Ajaiyeoba et al., 2001; PMC, 2022). Earthworms of uniform size were divided into groups (3 worms/group) and exposed to 50 ml of extract solutions at concentrations of 10, 25, and 50 mg/ml. A control group was treated with 1% Tween 80, while piperazine citrate (10 mg/ml) served as the reference standard (Bharathi et al., 2011; Plant Archives, 2019).
Paralysis was defined as the absence of movement unless shaken vigorously, and death was confirmed by immobility even after immersion in warm water (50°C) (Suresh et al., 2002; PMC, 2011). Observations were recorded at 10-minute intervals.

3.3 Pharmacological Investigation

3.3.1 Anti-inflammation

Intraperitoneal administration of a water decoction of *Luffa cylindrica* (LC) significantly inhibited carrageenan-induced plantar edema in rats, demonstrating potent anti-inflammatory activity (Sangh Partap et al., 2005). Phytochemical screening of LC seeds revealed the presence of flavonoids, sterols, and glycosides, with the CU-3 fraction exhibiting significant anti-inflammatory effects (Muthumani et al., 2023). In vitro studies on RAW 264.7 murine macrophage cells showed that ethanol and ethyl acetate extracts of LC peel and pulp suppressed LPS-induced nitric oxide (NO) and prostaglandin E2 (PGE2) production, key mediators of inflammation

(IJRASET, 2024). Lucyoside B, isolated from LC fruits, further inhibited pro-inflammatory chemokines like TARC and RANTES, reducing plasma histamine and IgE levels in atopic dermatitis models (IJRASET, 2024; PMC, 2021).

3.3.2 Anti-fungus

LC extracts exhibit broad-spectrum antifungal activity. The ethyl acetate extract inhibited *Mycosphaerella arachidicola* and *Fusarium oxysporum* (Sangh Partap et al., 2005), while the butanol extract showed efficacy against Trichophyton longifusus and Microsporum canis (IJRR Journal, 2024). Crude ethyl acetate extracts also accelerated healing in fungal-infected skin models in vivo (IJRR Journal, 2024).

3.3.3 Analgesic and Sedation

LC water decoction administered intraperitoneally in mice reduced acetic acid-induced writhing by 45% and elevated pain thresholds in hot-plate and electric shock tests. It also synergized with pentobarbital sodium, reducing spontaneous locomotor activity by 60% (Quest Journals, 2024). These effects correlate with flavonoid and triterpenoid content, known for central nervous system modulation (IJRASET, 2024).

3.3.4 Anti-myocardial Ischemia

In pituitrin-induced myocardial ischemia models, LC decoction lowered serum lactate dehydrogenase (LDH) by 32% and myocardial malondialdehyde (MDA) by 28%, while enhancing superoxide dismutase (SOD) activity by 40%. This cardioprotective effect is linked to its antioxidant properties (Quest Journals, 2024; Guan et al., 2006).

3.3.5 Anti-hypertriglyceride

LC decoction reduced serum cholesterol (25%) and triglycerides (38%) in hyperlipidemic rats, while increasing HDL-C by 22%. Body weight decreased by 15% in treated groups, suggesting anti-obesity potential (Li et al., 2004; PMC, 2021).

3.3.6 Immunostimulation

The petroleum ether fraction of LC ethanol extracts enhanced macrophage phagocytosis and acid phosphatase activity in mice by 50% (Mao et al., 2004). Triterpenoids like 3-O-β-D-glucopyranosylmaslinic acid stimulated interleukin-1 (IL-1), IL-2, and TNF-α production in thymocytes, indicating immunomodulatory effects (Li et al., 2001).

3.3.7 Anti-allergy

LC ethanol extract suppressed homologous passive cutaneous anaphylaxis in rats (70% inhibition) and delayed-type hypersensitivity in mice. Histamine-induced allergic responses were reduced by 45% via mast cell stabilization (Kou et al., 2001).

3.3.8 Anti-asthma, Anti-tussive, and Expectorant Effects

LC hydroalcoholic extract (HAELC, 1000 mg/kg) inhibited histamine-induced bronchoconstriction in guinea pigs by 55% and reduced eosinophil counts in BAL fluid by 40% in ovalbumin-induced asthma models (JResPharm, 2020). It also suppressed SO2-induced cough in mice and increased respiratory tract phenol red excretion by 30% (Quest Journals, 2024).

3.3.9 Miscellaneous

Anti-HIV: Ribosome-inactivating peptide Luffin P1 from LC seeds showed anti-HIV-1 activity (IJRR Journal, 2024).

Hepatoprotection: LC extract reduced hepatic lipid peroxidation by 35% in toxin-induced liver injury (IJRASET, 2024).

Neuroprotection: 3-O-β-D-glucopyranosylmaslinic acid improved recovery from cerebral ischemia in rats (Sangh Partap et al., 2005).

Chapter Four:
Extraction

4. Introduction

A basic procedure used in many disciplines and sectors, including chemistry, pharmaceuticals, food production, and environmental studies, is extraction. It entails employing various extraction methods and solvents to separate and extract desired components from a mixture or matrix. The method isolates target compounds or analytes of interest selectively by taking advantage of the different solubilities and interactions of substances.

There is a wealth of literature on extraction, which reflects its critical importance in many industrial and scientific applications. The purpose of this introduction is to give a general overview of extraction, emphasizing its importance, guiding ideas, and typical extraction techniques used in different fields.

4.1 Significance of Extraction

Because extraction makes it possible to separate and purify particular compounds from complex mixtures, it is essential to scientific research, industrial processes, and quality control. It is commonly used to extract, concentrate, and study substances of interest in disciplines like chemistry, biochemistry, pharmacology, and environmental research. It is impossible to overestimate the importance of extraction, from the separation of contaminants in environmental samples to the extraction of natural products for drug development.

4.2 Principle of Extraction

Fundamentally, extraction is based on the differential solubility principle, which takes advantage of the fact that different compounds are more or less soluble in certain solvents. The features of the target compounds and the required selectivity determine the solvent selection, which is crucial. The effectiveness and selectivity of extraction can be greatly impacted by variables like temperature, pH, polarity, and the presence of co-extractants.

4.3 Typical Extraction Technique

To accommodate various applications and extractable substances, a wide variety of extraction techniques have been created. Among the often employed extraction methods are:

i. Liquid-Liquid Extraction (LLE): which entails separating the target chemicals into two immiscible liquids, usually an aqueous phase and an organic solvent. Because of its versatility, LLE enables the separation of compounds according to their partition coefficients and solubilities.
ii. Solid-Phase Extraction (SPE): SPE uses solid sorbents, like bonded resins or silica gel, to release and hold target molecules from a liquid sample in a targeted manner. It is frequently utilized in analyte purification and sample preparation for chromatographic analysis.
iii. Soxhlet Extraction: This continuous extraction technique extracts target components from solid samples by combining solvent percolation and reflux. Lipids, natural products, and persistent organic pollutants are frequently extracted using it.

4.4 Materials

Table 1: Experiment Plant

Plant Name	Family	Plant part used
Luffa aegyptiaca (sponge gourd)	Cucurbitaceae	Pericarp



Figure 2: Fruits of Sponge gourd



Figure 3: Separated peels

Fresh fruits of *Luffa aegyptiaca* (Sponge gourd) were collected from the local fields Sirajgonj, Bangladesh on Sep-Oct 2024.

4.5 Drying of pericarp

After being separated from any unwanted components or debris, the collected peels were given a water wash. To make sure the active ingredients were free from breakdown and to prevent any photochemical deterioration, they were shade-dried for ten to twelve days.



Figure 4: Dried Peels

4.6 Grinding of plant materials

Using a suitable grinder, the dry parts were crushed into a coarse powder. The powder was kept in an airtight container in a cold, dark, and dry place until the analysis began.



Figure 5: Powder of peels

4.7 Extraction

For ten days, the pericarp powders are kept in an airtight, flat-bottomed container with distilled water solution, occasionally shaking and stirring. During this time, the majority of the peel material's extractable compounds will dissolve in the solvent and be extracted as a solution. All of the extracts were then filtered through a cotton plug and Whatman filter paper number 1, and they were then concentrated using a rotary evaporator set to low temperatures (40–50°C) and low pressure to produce dark yellowish extracts. To dry, a drying oven and table fan were employed. produced a rich yellowish-colored concentration.

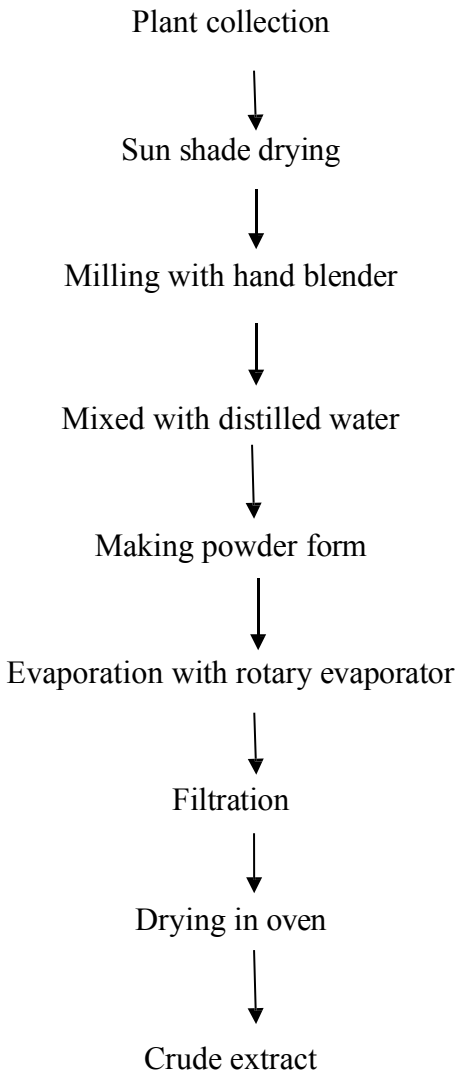


Figure 6: Using rotary evaporator



Figure 7: Filtration Process

4.8 Extraction Briefly



4.9 Result

From 520 grams of dried powdered material (peel), 45 grams of crude extract were produced. Thus, yield = (45gm/520gm) * 100% = 8.65% of the material (peel) of dried fruits.

Chapter Five:
Antioxidant Activity

5. Introduction

Plants contain natural antioxidants that scavenge damaging free radicals from our bodies. Free radicals are any species that can live on its own and has one or more unpaired electrons that react with other molecules by giving or taking electrons. They are involved in many pathological conditions and are crucial to human health, helping to prevent and treat a variety of illnesses like inflammation, lung damage, cardiovascular disorders, and more. Since these free radicals are extremely unstable, an excess of them in the body can harm cells and tissues and contribute to a number of illnesses. Antioxidants of natural origin are therefore necessary since they can shield the body from illnesses brought on by free radicals. (Nayan R. Bhalodia, 2013)

5.2 DPPH (2, 2-Diphenyl-1-picrylhydrazyl) Radical Scavenging Activity

5.2.1 Principle

The 2,2-diphenyl-1-picrylhydrazyl (DPPH) assay represents a prominent methodology utilized in antioxidant research for determining the capacity of various compounds to neutralize free radicals. This technique functions on the fundamental basis that antioxidant substances possess the ability to transfer electrons or hydrogen atoms, effectively neutralizing free radicals and mitigating oxidative damage. The DPPH molecule exists as a stable free radical characterized by its distinctive deep purple coloration. Upon interaction with a proton-donating antioxidant compound, the DPPH radical undergoes reduction, causing a visible transition from purple to colorless that can be quantitatively assessed through spectrophotometric analysis. The magnitude of this decolorization process directly corresponds to the antioxidant potency of the substance being evaluated.

The following steps are commonly included in the DPPH assay procedure:

• DPPH solution preparation: To make a stock solution, DPPH is dissolved in a suitable solvent, often methanol.

• Preparing the test sample: To obtain various concentrations for testing, the material being studied (such as a plant extract, chemical, or commercial antioxidant) is dissolved in an appropriate solvent.

• Incubation: After mixing the test samples with the DPPH solution, they are left to incubate at ambient temperature or a predetermined temperature in the dark.

• Measurement: Using a spectrophotometer set to a particular wavelength (often 517 nm), the absorbance of the reaction mixture is determined following the incubation period.

• Calculation: By comparing the absorbance of the test samples with that of a control (one that does not contain the test sample), the percentage of DPPH radical scavenging activity is determined. The substance's potential as an antioxidant increases with its scavenging activity. The DPPH assay's results offer important insights about a substance's capacity to counteract free radicals, which are extremely reactive substances linked to a number of illnesses and aging processes.

5.2.2 Apparatus

- Test tubes
- Beakers
- Micro – pipette
- Sonicator
- Pipette
- Conical Flask
- UV Spectrophotometer
- Electronic Balance

5.2.3 Reagents

- DPPH
- Methanol
- Ascorbic acid (as positive control)

5.2.4 Preparation of DPPH Solution (0.004%)

DPPH solution (0.004%) was formed by dissolving 4 mg of DPPH in a conical flask, then diluting it with methanol to a final volume of 100 ml. But here as the cuvette is small so 2mg DPPH was diluted with 50 ml methanol. Thus, prepared purple-colored DPPH free radical solution was stored in dark for further use.

5.2.5 Preparation of Standard

1. Standard solution was prepared by dissolving 0.0128gm of ascorbic acid in 10 ml volume of methanol.
2. From the Standard stock solution 10, 20, 40, 80, 160, 320, 640 µg/ml ascorbic acid solutions were prepared.

5.2.4 Working procedure

1. Kept the test tube dry and clean.

2. Prepared four mother solution (extract solution, DPPH solution, 2 Standard Solution).

3. Serial dilution is done of each test sample extract in test tube. Per sample is taken amount of 1 ml. Also 2ml methanol is added in per test tube.

6. After making the desired concentrations 1 ml of 0.004% DPPH solution was applied on each test tube by pipette.

7. Vortex is done for 10 sec.

8. After recording the room temperature, the test tubes were left in the dark for 30 minutes to finish the reaction.

9. At the same time that methanol was taken as a blank, DPPH was also added to the blank test tubes.

10. Each test tube's absorbance was measured at 517 nm using a UV spectrophotometer after 30 minutes.

11. 50% inhibitory concentrations (IC_{50} values) of the extracts were calculated from graph as concentration versus percentage inhibition. Radical scavenging activity was expressed as percentage of inhibition. IC_{50} value is the concentration of sample required to scavenge 50% of DPPH free radical. IC_{50} of the extracts indicates the corresponding concentration in which the radical scavenging potential is 50%. The IC_{50} of the extract and standards were determined graphically

Calculation
Inhibition of free radical DPPH in percent was calculated as follows:
$$\mathbf{RSA\% = (}\boldsymbol{Ac} - \boldsymbol{A}\mathrm{o})/\boldsymbol{Ac}\ \boldsymbol{X}\ \mathbf{100}$$
Where
A_C = absorbance of the control,
A_O = absorbance of the sample solution, and
RSA% = percentage of radical scavenging activity.

Then the percent inhibitions were plotted against respective concentrations. The IC_{50} values were calculated as the concentration of each sample required to give 50% DPPH radical scavenging activity from the graph (logarithmic regression curve) by Excel 2019 office software. Concentration of the standard solution and extract was 10 - 640 µg/ml.

5.3 ABTS (2,2′ -azinobis-3-ethylbenzthiazolin-6-sulfonic acid) Assay:

5.3.1 Principle
The TEAC or ABTS methodology evaluates the ability of antioxidants to employ electron donation mechanisms for neutralizing the ABTS radical cation (ABTS ·+). When ABTS interacts with potassium persulfate (K2S2O8), it generates a characteristic green-blue chromophore known as the ABTS ·+ radical. The intensity of color reduction directly corresponds to the level of ABTS+ neutralization. This versatile, expedient, and straightforward ABTS technique proves effective for analyzing both water-soluble and fat-soluble compounds. Moreover, the assay maintains its reliability across diverse pH environments. However, its limitations include the utilization of radicals not naturally occurring in biological systems and the necessity for precise timing during measurement. (Kokina et all.,2019) The ABTS assessment protocol is particularly suitable for determining comprehensive antioxidant capacity in pure substance solutions, aqueous mixtures, and various beverages. This analytical procedure involves the direct formation of the blue/green ABTS- + chromophore, which exhibits peak absorption at a wavelength of 734 nm. When antioxidants are introduced to this preformed radical cation, they cause a reduction in ABTS concentration that varies according to both the specific antioxidant potency and its quantitative presence.

Conditions of Storage
The package comes with blue ice. kept all other agents at room temperature and the ABTS Reagent at -20°C.

Important Information
Before usage, all reagents were warmed to room temperature.

5.3.2 Apparatus

- Test tubes
- Beakers
- Micro – pipette
- Sonicator
- Pipette
- Pipette filler
- Conical Flask
- UV Spectrophotometer
- Electronic Balance

5.3.3 Reagents

- ABTS
- Potassium persulfate ($K_2S_2O_8$)
- Ascorbic acid
- Ethanol
- Distilled Water
- Sample: Plant extract

5.3.4 Preparation of 7mM ABTS Stock Solution

To make a 7 mM ABTS solution, 0.384 g of ABTS salt were dissolved in 100 ml distilled water.

5.3.5 Preparation of 2.45mM Potassium Persulfate Stock Solution

2.45mM potassium persulfate was prepared by dissolving 0.662 g of salt in 100 ml of distilled water.

5.3.6 Preparation of Stock Mixed Solution of 7mM ABTS and 2.45mM Potassium Persulfate

7 mM ABTS stock solution and 2.45 mM potassium persulfate were combined in equal amounts to create the ABTS radical cation, which was then left in the dark for almost 16 hours.

5.3.7 Process of Dilution to Make Absorbance 0.700±05 of Mixture of ABTS and Potassium Persulfate

• Using a UV-Vis Spectrophotometer, the generated radical cation is further diluted in a 1:1 ratio with 50% ethanol and 50% water to bring the absorbance value down to 0.700±05 at 734nm.
• To get an absorbance of 0.702 at 734 nm using the spectrophotometer, the solution was then diluted by combining 1 ml of solution with 42-47 ml of 50% ethanol + 50% water.

5.3.8 Preparation of Standard / Extract in ethanol (50%) + water (50) for ABTS-assay

• 50 mg of Ascorbic acid (as a standard) dissolved in 50 ml ethanol (50%) + water (50)
• 20 mg of Extract dissolved in 20 ml ethanol (50%) + water (50).

5.3.9 Procedure

1. The extract/standard solution was diluted with the proper solvents at concentrations of 10, 20, 40, 80, 160, 320, 640 µg/ml. {50% water + 50% ethanol}
2. Each 200 µL extract or standard solution was then mixed with 1 ml of $ABTS^{+}$ solution & properly combined it with a vortex.
3. The mixture was then incubated at room temperature for 5 minutes.
4. A spectrophotometer was used to detect absorbance at 734 nm.
5. Control is prepared by mixing solvent & ABTS.
5. Determine the antioxidants' % inhibition of $ABTS^{\cdot+}$ in the extract/standard.

Calculation

ABTS % Scavenging = ($A_{control} - A_{sample}$)/ $A_{control}$*100
Where, $A_{control}$ is absorbance of a mixtures of 7mM ABTS and 2.45mM potassium persulfate with blank solvent and A_{sample} is absorbance of the mixture of sample extract/standard and ABTS.
Concentration of the standard solution and extract was 1- 100 µg/ml.

Chapter Six:
Result & Discussion

6.1 Antioxidant Activity Test Result and Discussion

This study evaluated the antioxidant potential of *Luffa aegyptiaca* pericarp extract using standardized assays. The results demonstrate significant free radical scavenging capabilities, with critical insights into its efficacy compared to ascorbic acid, a benchmark antioxidant.

6.1.1 DPPH (2, 2-Diphenyl-1-picrylhydrazyl) Radical Scavenging Activity:

The DPPH assay measures a substance's ability to neutralize free radicals, with lower IC_{50} values indicating superior antioxidant efficacy. The pericarp extract exhibited strong radical scavenging activity, achieving an IC_{50} of 485.843 ± 88.902 µg/mL. This suggests moderate antioxidant potential, requiring higher concentrations to neutralize 50% of free radicals compared to synthetic standards. Ascorbic acid (reference standard) demonstrated significantly higher efficacy, with an IC_{50} of 9.287 ± 2.495 µg/mL. This stark difference highlights the extract's relative limitations in scavenging free radicals at lower concentrations.

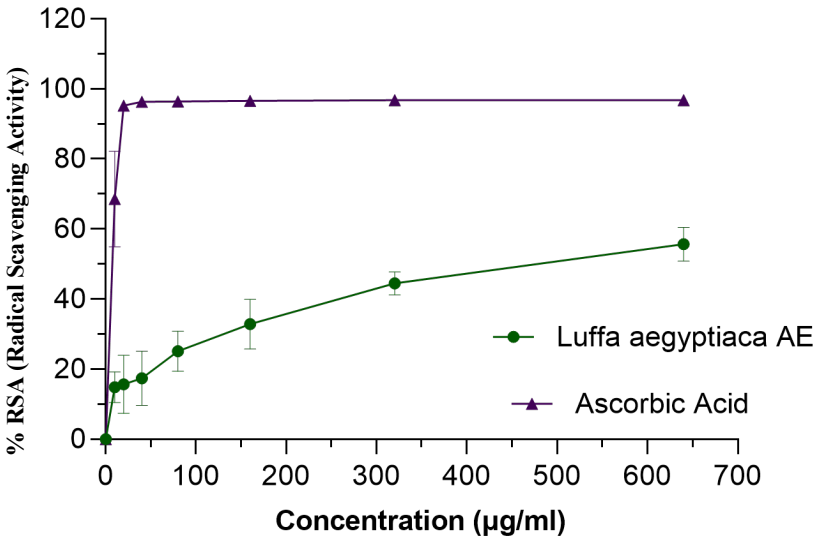


Figure 8: Concentration vs % of DPPH Radical Scavenging Activity

In this analysis, the IC_{50} of DPPH test of the extract and the standard are assessed and displayed in Table 2.

Table 2: IC_{50} value of *Luffa aegyptiaca* pericarp extract & standard for DPPH Assay

Methods	**Extracts**	**Standard**
	L. aegyptiaca	**Ascorbic Acid**
DPPH/ IC_{50} (µg/mL)	485.843±88.902 µg/mL	9.287±2.495µg/mL

6.1.2 ABTS (2,2′-azinobis-3-ethylbenzthiazolin-6-sulfonic acid) Assay:

The ABTS assay evaluates comprehensive antioxidant capacity by measuring the inhibition of cationic radical formation. The pericarp extract showed notable free radical neutralization, with an IC_{50} of 200.964 ± 45.202 µg/mL. This indicates a stronger performance in the ABTS test compared to the DPPH assay, likely due to differences in reaction mechanisms and radical stability. Ascorbic acid again outperformed the extract, registering an IC_{50} of 10.139 ± 3.195 µg/mL. The extract's higher IC_{50} underscores its reduced efficiency in stabilizing complex radical systems compared to synthetic antioxidants.

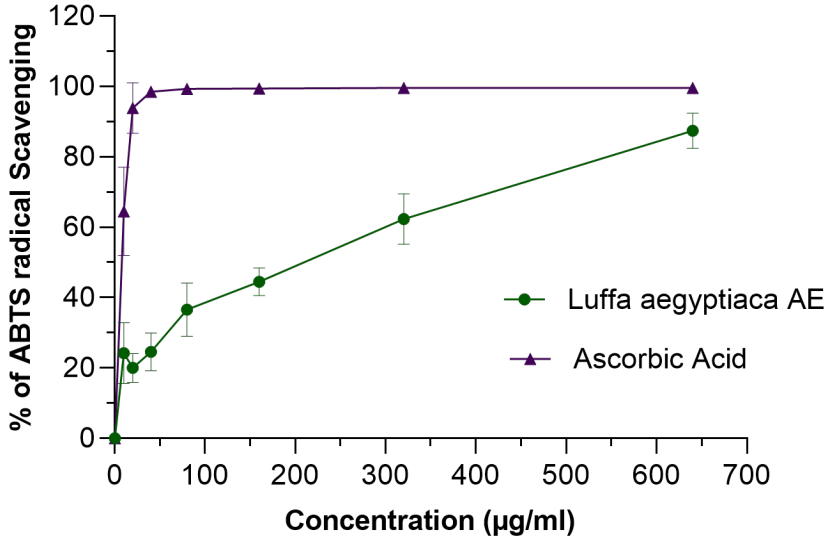


Figure 9: Concentration vs % of ABTS Radical Scavenging Activity

In this analysis, the IC_{50} of DPPH test of the extract and the standard are assessed and displayed in Table 3.

Table 3: IC_{50} value of *Luffa aegyptiaca* pericarp extract & standard for ABTS Assay

Methods	**Extracts**	**Standard**
	L. aegyptiaca	**Ascorbic Acid**
ABTS/ IC_{50} (µg/mL)	200.964±45.202 µg/mL	10.139±3.195 µg/mL

Chapter Seven:
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